

IT549: Deep Learning

Lab Assignment 4

Object Detection Evolution: From R-CNN to YOLO

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1 Assignment Overview

This assignment explores the evolution of object detection models, starting from early region-proposal methods such as R-CNN and progressing toward modern architectures such as Faster R-CNN and YOLO.

The objective of this lab is to understand the architectural improvements that significantly improved detection speed and efficiency.

The dataset used for this assignment is the Fruit Images dataset containing three object classes:

- Apple
- Banana
- Orange

Each image includes bounding box annotations provided in normalized and pixel coordinate formats.

The tasks implemented in this assignment follow the historical progression of object detection methods.

This assignment specification is provided in the lab instructions. :contentReference[oaicite:0]index=0

2 Dataset

The dataset used in this assignment is the Fruit Images dataset from Kaggle.

The dataset contains images of fruits with bounding box annotations corresponding to each object instance.

Each annotation includes:

- Object class label
- Bounding box coordinates
- Normalized and pixel-based formats

The dataset is used for both visualization and model evaluation throughout the assignment.

3 Task 0: Ground Truth Visualization

A random image from the dataset is loaded and its annotation file is parsed.

Bounding boxes are extracted from the annotation file and drawn on the image using OpenCV and Matplotlib.

This step verifies that the dataset annotations are correctly interpreted before running object detection models.

4 Task 1: Intersection over Union (IoU)

Intersection over Union (IoU) measures the overlap between two bounding boxes and is defined as:

$$IoU = \frac{Area\ of\ Intersection}{Area\ of\ Union}$$

A Python function was implemented from scratch to compute IoU between two bounding boxes in the format:

$$[x_{min}, y_{min}, x_{max}, y_{max}]$$

The function was tested using three example scenarios:

- Highly overlapping boxes
- Partially overlapping boxes
- Completely disjoint boxes

5 Task 2: Selective Search Region Proposals

Selective Search is used to generate candidate object regions.

Using OpenCV's SelectiveSearchSegmentation module, the top 200 region proposals were generated for a random image.

These region proposals were visualized by overlaying bounding boxes on the original image.

This step replicates the first stage of the original R-CNN pipeline.

6 Task 3: R-CNN Feature Extraction Bottleneck

In the R-CNN pipeline, each region proposal is cropped and passed through a convolutional neural network individually.

A pre-trained ResNet18 model from torchvision was used with the final classification layer removed.

For the top 100 region proposals:

- Each region was cropped from the image
- Resized to 224x224

- Passed through the ResNet18 model to extract features

The execution time required to process these crops was recorded.

This demonstrates the major computational bottleneck of the original R-CNN architecture.

7 Task 4: Fast R-CNN with RoI Pooling

Fast R-CNN improves efficiency by running the CNN only once on the entire image.

The resulting feature map is used to extract region features using RoI pooling.

Steps implemented:

- Pass the full image through ResNet convolution layers
- Generate the feature map
- Use `torchvision.ops.roi_pool`
- Extract features for the same region proposals

Execution time was recorded and compared to Task 3.

7.1 Conceptual Analysis

Fast R-CNN eliminates redundant convolution operations by computing the convolutional feature map only once for the entire image instead of repeatedly processing each region proposal independently.

This significantly reduces computation time and improves detection efficiency.

8 Task 5: Faster R-CNN

Faster R-CNN integrates the region proposal mechanism directly into the network using a Region Proposal Network (RPN).

A pre-trained `fasterrcnn_resnet50_fpn` model from torchvision was used.

Steps performed:

- Load the pre-trained model
- Run inference on a test image
- Extract predicted bounding boxes, labels, and confidence scores
- Filter predictions with confidence score below 0.80
- Visualize the final detections

8.1 Conceptual Analysis

The Region Proposal Network generates candidate object regions directly from the feature maps produced by the backbone CNN.

Because the RPN shares convolutional features with the detection network, Faster R-CNN removes the need for external algorithms such as Selective Search.

This results in faster and more efficient object detection.

9 Task 6: Non-Maximum Suppression (NMS)

Non-Maximum Suppression is used to remove redundant bounding box predictions.

The algorithm implemented in this assignment performs the following steps:

1. Sort bounding boxes by confidence score
2. Select the highest scoring box
3. Compute IoU with remaining boxes
4. Remove boxes with IoU greater than the threshold
5. Repeat until no boxes remain

9.1 Conceptual Analysis

The IoU threshold controls how aggressively duplicate detections are removed.

A high threshold (e.g., 0.9) allows many overlapping boxes to remain, which may produce multiple detections for the same object.

A low threshold (e.g., 0.1) aggressively removes overlapping boxes, which may suppress valid detections when objects are close together.

For tightly packed objects, a moderately high threshold is typically preferred.

10 Task 7: YOLO Fine-Tuning

YOLO reformulates object detection as a single-stage regression problem.

The Ultralytics implementation of YOLOv8 was used.

Steps performed:

- Install the ultralytics package
- Load the pre-trained YOLOv8 Nano model
- Fine-tune the model on the fruit detection dataset
- Train the model for 10 epochs
- Evaluate model performance

Inference was performed on several test images and predictions were visualized.

11 Model Comparison

Model	Inference Time (ms)	Precision	Recall
Faster R-CNN (pretrained)	152.5	0.448	0.896
YOLOv8 (pretrained)	63.6	0.266	0.129
YOLOv8 (fine-tuned)	24.4	0.873	0.829

12 Conclusion

This lab demonstrated the evolution of object detection architectures from R-CNN to modern single-stage detectors like YOLO.

Key observations include:

- Early object detection models relied on computationally expensive region proposals.
- Fast R-CNN improved efficiency using shared feature maps.
- Faster R-CNN introduced the Region Proposal Network.
- YOLO further simplified detection by treating it as a single-stage regression task.

These architectural innovations significantly improved object detection speed and performance.